

IDENTIFYING INFORMATION:

NAME: Kalyanapu, Alfred

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-7746-3121>

POSITION TITLE: Professor

PRIMARY ORGANIZATION AND LOCATION: Tennessee Technological University, Civil and Environmental Engineering, Cookeville, Tennessee, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Utah, Salt Lake City, Utah, USA	PHD	08/2007 - 08/2011	Civil Engineering
University of Utah, Salt Lake City, Utah, USA	MS	08/2004 - 05/2007	Civil Engineering
National Institute of Technology, Warangal, Warangal, Telangana, India	BOTH	06/1999 - 04/2003	Civil Engineering

Appointments and Positions

2023 - present	Professor, Tennessee Technological University, Civil and Environmental Engineering, Cookeville, Tennessee, United States
2025 - present	Interim Chair, Department of Electrical and Computer Engineering, Cookeville, Tennessee, United States
2022 - 2023	Interim Co-Director, Tennessee Technological University, Center for the Management, Utilization and Protection of Water Resources, Cookeville, Tennessee, United States
2017 - 2023	Associate Professor, Tennessee Technological University, Department of Civil and Environmental Engineering, Cookeville, TN, United States
2011 - 2017	Assistant Professor, Tennessee Technological University, Department of Civil and Environmental Engineering, Cookeville, TN, United States
2007 - 2011	Graduate Research Assistant, University of Utah, Department of Civil and Environmental Engineering, Salt Lake City, UT, USA
2006 - 2007	Graduate Research Assistant, Los Alamos National Laboratory, Los Alamos, NM, USA
2004 - 2006	Graduate Research Assistant, University of Utah, Department of Civil & Environmental Engineering, Salt Lake City, UT, USA

Products

1. Darkwah G, Kalyanapu A, Owusu C. Machine Learning-Based Flood Forecasting System for Window Cliffs State Natural Area, Tennessee. GeoHazards. 2024; 5(1):64-90. issn: 2624-795X
2. Gangrade S, Ghimire G, Kao S, Morales-Hernández M, Tavakoly A, Gutenson J, Sparrow K, Darkwah G, Kalyanapu A, Follum M. Unraveling the 2021 Central Tennessee flood event using

- a hierarchical multi-model inundation modeling framework. *Journal of Hydrology*. 2023; 625:130157. issn: 0022-1694
3. Kalyanapu A, Owusu C, Wright T, Datta T. Low-Cost Real-Time Water Level Monitoring Network for Falling Water River Watershed: A Case Study. *Geosciences*. 2023; 13(3):65. issn: 2076-3263
 4. Sharif M, Hines T, Ghafoor S, Morales-Hernández M, Dullo T, Kalyanapu A. A Performance Prediction Model for Structured Grid Based Applications in HPC Environments. 2023 22nd International Symposium on Parallel and Distributed Computing (ISPDC). 2023; :77-84. isbn: 9798350341270
 5. Sava E, Cervone G, Kalyanapu A. Multiscale Observation Product (MOP) for Temporal Flood Inundation Mapping of the 2015 Dallas Texas Flood. *Remote Sensing*. 2023; 15(6):1615. issn: 2072-4292
 6. Latham J, Dadzie G, Kalyanapu A, Shannigrahi S. AquaCam: An ML-Enhanced Low-Cost, Deploy-Anywhere Water Level Detection Sensor. 2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops). 2025; :628-632. isbn: 9798331535537
 7. Fordjour G, Nur F, Kalyanapu A, Cervone G. Enhancing flood water level prediction through transfer learning with warmup learning rate scheduling in LSTM models: a comparative case study in Kentucky and Tennessee watersheds. *Modeling Earth Systems and Environment*. 2025; 11(1):14. issn: 2363-6203
 8. Kao S, Kalyanapu A, Ghafoor S, Evans K. High-performance computing in water resources hydrodynamics. *Journal of Hydroinformatics*. 2020; 22(5):1217-1235. issn: 1464-7141
 9. Dullo T, Gangrade S, Morales-Hernández M, Sharif M, Kalyanapu A, Kao S, Ghafoor S, Ashfaq M. Assessing Climate Change-Induced Flood Risk in the Conasauga River Watershed: An Application of Ensemble Hydrodynamic Inundation Modeling. *Natural Hazards and Earth System Sciences Discussions*. 2020; 2020:1-54. issn: 2195-9269
 10. Owusu C, Masto N, Kalyanapu A, Murdock J, Cohen B. Shallow water depth estimation of inland wetlands using Landsat 8 satellite images. *Remote Sensing*. 2024; 16(16):2986. issn: 2072-4292

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment

program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Kalyanapu, Alfred in SciENCv on 2026-04-19 22:03:38